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1. A free-rider problem arises when a good is

(A) nonrival
(B) nondepletable
(C) nonexcludable
(D) produced in a competitive market
(E) produced in a monopolistic market

2. An increase in the marginal social benefit of consuming a public good should result in

(A) a decrease in public production of the good
(B) an increase in the optimal quantity of the good
(C) a decrease in the optimal quantity of the good
(D) a decrease in the social costs associated with producing the good
(E) a decrease in the social costs associated with consuming the good

3. A pure public good is a good that is

(A) provided efficiently by markets
(B) rivalrous and excludable in consumption
(C) nonrivalrous and excludable in consumption
(D) rivalrous and nonexcludable in consumption
(E) nonrivalrous and nonexcludable in consumption

4. Assuming that all residents of the city of Smallville prefer to have streetlights installed citywide, which of the following best explains why individual residents do **not** install the optimal number of streetlights?

(A) Streetlights generate negative externalities.
(B) Streetlights are nonexcludable.
(C) Installing streetlights results in diseconomies of scale.
(D) The marginal cost of installing streetlights is zero.
(E) The marginal benefit of installing streetlights is zero.

5. Characteristics of a pure public good include which of the following?

I. Nonpaying consumers cannot be excluded from consuming the good.

II. Consumption by any one person does not reduce the availability of the good to others.

III. The amount consumed depends on the price paid.

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) II and III only

6. An example of a good that is nonrival and nonexcludable is

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(A) clean water in a stream
(B) a bicycle
(C) higher education
(D) a lighthouse
(E) an entertainment website requiring a paid subscription

7. One major difference between public goods and private goods is that
(A) public goods exclude nonpaying consumers, while private goods are available to all
(B) public goods are inferior goods, while private goods are normal goods
(C) public goods generate negative externalities, while private goods generate positive externalities
(D) it is difficult to charge customers directly for the provision of public goods
(E) it is impossible for private firms to produce public goods

8. One reason many governments provide free access to mandatory primary education is because primary education
(A) generates negative externalities
(B) helps to reduce future income inequality
(C) is only provided by a natural monopoly
(D) is nonrival and therefore subject to overcrowding
(E) is nonexcludable and therefore subject to the free rider problem

9. Public goods are underproduced in a competitive market because the free-rider problem causes which of the following?
(A) Fixed supply
(B) Effective price ceilings
(C) Reduced market demand
(D) Negative externalities
(E) Increased imports

10. Which of the following is the best example of the free-rider problem?
(A) A company gives away free samples of a new product.
(B) A single firm is the sole supplier of corn oil.
(C) There are economies of scale in the production of passenger elevators.
(D) It is difficult to exclude those who do not pay for a good from enjoying the benefits of the good.
(E) There is free entry and exit into production of toothpaste, but products are not homogeneous.

11. Private supply of public goods is most likely to result in
(A) less than the efficient level of output, due to the free-rider problem
(B) less than the efficient level of output, due to the problem of insufficient competition
(C) more than the efficient level of output, due to lower costs of private firms
(D) more than the efficient level of output, due to the existence of externalities
(E) an overuse of these goods

12. The government must provide public goods such as national defense because

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(A) the production of public goods requires economies of scale that the private sector cannot achieve
(B) it is generally impossible to exclude individuals who value public goods but do not pay for them
(C) public goods cannot be produced in private competitive markets, since they have highly inelastic demand
(D) private producers charge a price that is substantially greater than marginal cost
(E) no single individual should have to pay for public goods, since they benefit society as a whole

13. Which of the following explains why free-riding can result in a market failure?
(A) More than the socially optimal quantity is produced and consumed.
(B) The socially optimal quantity is produced but less than the socially optimal quantity is consumed.
(C) Private producers of nonexcludable goods are unable to charge everyone who consumes the good.
(D) There is a market surplus of the good.
(E) There is no consumer surplus derived from the good.

14. Which of the following is true of the marginal cost of providing a pure public good to one more consumer?
(A) It is positive.
(B) It is equal to zero.
(C) It is equal to the original cost of the good.
(D) It decreases as the number of consumers increases.
(E) It increases as the number of consumers increases.

15. National defense is generally provided by the public sector for which of the following reasons?
(A) The opportunity cost of privately providing national defense is too high.
(B) The government always provides the socially optimal quantity.
(C) National defense is a natural monopoly.
(D) National defense is subject to negative externalities.
(E) The benefits of national defense are mostly nonrival and nonexcludable.

16. National defense is a good example of a pure public good because it is
(A) nonrival and nonexcludable
(B) nonrival and excludable
(C) rival and nonexcludable
(D) rival and excludable
(E) free and provided at zero cost

17. National defense is an example of a public good because
(A) it requires tax revenues to fund any production
(B) one person's use of it will decrease another person's ability to use it
(C) it is nonexcludable and nonrival
(D) the private market typically produces the socially efficient level of output
(E) the public is protected from invasion

18. Which of the following is the best example of a pure public good?

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(A) Electricity from a public utility
(B) Mail delivery service by the post office
(C) Social Security payments
(D) National defense
(E) Imported oil

19. A free-rider problem exist when
(A) a good is excludable in consumption.
(B) a good is nonexcludable in consumption.
(C) a good is rival in consumption.
(D) a good is nonrival in consumption.
(E) a good is in abundant supply.

20. Which of the following is an example of a good that is nonexcludable and rival in consumption?
(A) Fish sandwich
(B) Public highway
(C) Private dance lesson
(D) Online music
(E) National defense

21. Which of the following best defines a pure public good?
(A) A good that is provided free of charge to the public
(B) A good that is provided by the government, whether or not a fee is charged
(C) A good that is efficiently produced only by a monopoly
(D) A good whose benefits can be enjoyed by many consumers at the same time and from which consumers cannot be excluded
(E) A good that by law must be available to all consumers without discrimination on the basis of sex, race, or religion

22. Good X, a popular item with consumers, is nonrival and non-excludable in consumption. What is the most likely outcome in the market for Good X if there is no government intervention?
(A) Firms will provide the socially optimal quantity of Good X because consumers will have a demand for it.
(B) Firms will provide the socially optimal quantity of Good X because they will be profitable in the short run.
(C) Firms will provide the socially optimal quantity of Good X despite the fact that marginal social cost is greater than the marginal private cost at the equilibrium market quantity.
(D) Firms will not provide the socially optimal quantity of Good X because people are able to consume it without having to pay for it.
(E) Firms will not provide the socially optimal quantity of Good X because the marginal social cost is greater than the marginal private cost at the equilibrium market quantity.

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23. The city council divides a community's residents into three groups: individual young adults, families with children, and older adults. The following table summarizes how much each group is willing to pay for each playground.

	Amount Groups Are Willing to Pay for Each Playground		
Number of Playgrounds	Individual Young Adults	Families with Children	Older Adults
1	$400	$2,000	$1,000
2	$300	$1,800	$900
3	$200	$1,600	$800
4	$100	$1,400	$700
5	$0	$1,200	$600
6	$0	$1,000	$500
7	$0	$900	$400
8	$0	$800	$300

The city council must pay $2,250 to build each playground. Which of the following is a characteristic of playgrounds and what is the optimal number of playgrounds for the township to build?

(A) Playgrounds are nonrival in consumption, and the optimal number of playgrounds is zero.

(B) Playgrounds are nonrival in consumption, and the optimal number of playgrounds is two.

(C) Playgrounds are rival in consumption, and the optimal number of playgrounds is three.

(D) Playgrounds are nonexcludable, and the optimal number of playgrounds is zero.

(E) Playgrounds are excludable in consumption, and the optimal number of playgrounds is two.

24. Which of the following is a characteristic of a public good?

(A) It is excludable.

(B) It is nonrival in consumption.

(C) It yields substantial spillover costs.

(D) The private sector would produce a larger than socially optimal quantity.

(E) Private-sector producers would realize too great a profit.

25. Which of the following is an example of a pure public good?

(A) A hamburger sold in a restaurant on a college campus

(B) Medical education to become a doctor

(C) A hurricane warning siren

(D) Paper pulp produced with pollution emissions

(E) A ticket to the movie theater

26. Public goods are under produced in a competitive market because the free-rider problem causes

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(A) supply to be perfectly elastic
(B) a price ceiling to become binding
(C) consumers to not reveal their true demand
(D) positive externalities to increase
(E) demand to be perfectly inelastic

27. Public goods are typically provided by the government, because private markets tend to
(A) produce them at higher cost
(B) produce less than the socially optimum quantity
(C) produce more than the socially optimum quantity
(D) charge a monopoly price that is greater than the marginal social cost
(E) charge a monopoly price that is less than the marginal social cost

28. Which pair of characteristics is associated with a pure public good?
(A) Nonrival consumption and ability to exclude nonpayers
(B) Increasing marginal utility of consumption and a free-rider problem
(C) Rival consumption and an ability to exclude nonpayers
(D) Rival consumption and a free-rider problem
(E) Nonrival consumption and a free-rider problem

29. If a pure public good is provided only by a private market, then generally the good will
(A) be overproduced relative to the socially optimal quantity
(B) be underproduced relative to the socially optimal quantity
(C) be excludable
(D) be rivalrous
(E) reduce deadweight loss

30. Which of the following best demonstrates nonrivalry in consumption?
(A) Occupying a seat in a sold-out movie theatre
(B) Sharing rent equally among roommates
(C) Catching fish in a local pond
(D) Playing golf on a crowded public golf course
(E) Listening to a radio program

31. Tornado-warning sirens are usually paid for by local governments rather than private individuals because
(A) each individual does not find the sirens to be of value
(B) sirens are rival and excludable in consumption
(C) private individuals lack the incentive to cover the cost of providing the sirens
(D) sirens will be overproduced in the market equilibrium
(E) it is cheaper to produce sirens for local governments than to produce them for private individuals

32. Assume that many boats have free and open access to fishing in the waters of a country. Which of the following statements is true?

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(A) With the availability of fish for all consumers, there is a positive externality; therefore, governments need to encourage greater use of the common areas with more production and increased consumption.

(B) Over time, fishing operations will voluntarily leave the market as a result of increased competition.

(C) There will be a tendency for fishing operations to overfish, reduce the stock of fish, and limit the future availability of fish for succeeding generations.

(D) The waters of a country are a rare example of a resource that is not scarce. Therefore, standard economic analysis does not apply.

(E) The advantage of allowing free and open access to fishing is to limit the harm from a negative externality where too few fish are currently caught.

33. Which of the following best explains why the government provides public goods?

(A) Nonexcludable goods have a perfectly inelastic demand.

(B) Nonrival goods benefit only private consumers.

(C) Surpluses result from the private production of nonrival goods.

(D) Production of socially optimal quantities results in positive economic profits.

(E) Nonexcludable goods are underproduced by private firms.